

Master Curriculum and Operations Guide

Two camps, 24 primary missions and 8 backups, using activity-specific sequences to predict, build, observe, measure, solve, score, and defend. Redesign appears only where the method and rubric include it.

EXECUTIVE SUMMARY

The program runs as two distinct STEM seasons for middle school students: From Trees to Tech with 16 students in 4 teams of 4, and PY-STEM with up to 32 students. From Trees to Tech treats nature as the engineer: trees, leaves, soil, water, ecosystems, sensors, and climate resilience. PY-STEM is broad applied STEM: biomedical engineering, waves, optics, mechanics, materials, accessibility design, and algorithms, with light tie-ins to the Science History Institute and the Charles Library BookBot.

Every primary activity has a measurable win condition and a 100-point rubric. Competition is built into the science task itself, not added on top. The two camps share no core mechanic, build type, or scoring loop.

FROM TREES TO TECH: 12 PRIMARY ACTIVITIES

ID	ACTIVITY	CORE IDEA AND WIN
TTT-01	Mud Battery Bioelectric League (Biology + Energy)	Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.
TTT-02	Forest Sensor Sprint (Field Science)	Best evidence-based sensor recommendation with accurate measurements and efficient route planning.
TTT-03	Seed Dispersal Derby (Biomimicry)	Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.
TTT-04	Xylem Pipeline Relay (Biomimicry)	Most water delivered with speed, low spill, and strong explanation.
TTT-05	Greenhouse Climate Controller (Ecology)	Best plant-zone-control match supported by tour evidence.
TTT-06	Pinecone Weather Machine (Biomimicry)	Most accurate, readable, and fast response after spray-and-dry trials.
TTT-07	Pollinator Network Draft and Build (Ecology)	Highest pollinator support across spring, summer, and fall using native and clumping logic.
TTT-08	Arboretum Eco-Quest (Field Science)	Most accurate checkpoint solutions with safe field conduct.
TTT-09	Minecraft Tree World Resilience Cup (Ecology)	Best climate-resilient design with accurate nature-based strategies.

ID	ACTIVITY	CORE IDEA AND WIN
TTT-10	Tree Ring Climate Detective (Data + Climate)	Most accurate use of the practice-card code, supported with claim-evidence-reasoning.
TTT-11	Lotus Leaf Surface Sprint (Materials)	Best clean runoff and least residue after one redesign.
TTT-12	Leaf Stomata Microscope Detective (Field Science)	Most reliable density comparison with mean, range, and an evidence-limited hypothesis.

PY-STEM: 12 PRIMARY ACTIVITIES

ID	ACTIVITY	CORE IDEA AND WIN
PYS-01	Magnetic Capsule Maze Cup (Biomedical)	Fastest clean run with low penalties and a clear magnetic-control explanation.
PYS-02	Oobleck Armor Arena (Materials)	Best target protection with least material and cleanest reset.
PYS-03	Cardboard Automata Arcade (Mechanics)	Most reliable motion with useful mechanism and clear explanation.
PYS-04	Stethoscope Sprint and Recovery (Biomedical)	Best sound quality plus accurate, respectful data collection.
PYS-05	Reaction Time Combine (Biomedical)	Best median plus most improvement using clean data.
PYS-06	SONAR Slinky Showdown (Waves + Sound)	Most correct wave predictions and station solutions.
PYS-07	Pinhole Precision Challenge (Optics + Light)	Best image clarity and concept explanation after one aperture redesign.
PYS-08	Low-Ropes Force Map Relay (Mechanics)	Most accurate predictions and clear debrief connections.
PYS-09	Hovercraft Hockey Hackathon (Mechanics)	Highest normalized glide and five-shot target scores, with explanation and build quality.
PYS-10	Spectra Sleuth Showdown (Optics + Light)	Most correct spectrum matches and strongest safe fireworks-science explanation.
PYS-11	BookBot Bin Logic Challenge (Computing)	Most correct retrievals with the fewest legal moves, card-normalized efficiency, and clear algorithm defense.
PYS-12	Accessibility Ramp Rescue Lab (Design)	Best safe ramp meeting slope, load, portability, and client criteria.

BACKUP ACTIVITIES

Held ready for weather, trip delays, or materials failure. Activity-specific sequence, lighter setup.

ID	ACTIVITY	CORE IDEA AND WIN
TTB-01	Root Grip Erosion Rescue (Field Science)	Most soil retained after standard pour.
TTB-02	Tree Height Triangulation Shootout (Field Science)	Closest to instructor reference height.
TTB-03	Urban Heat Shade Dash (Data + Climate)	Best data-backed cool route.
TTB-04	Photosynthesis Float-Off Playoffs (Field Science)	Fastest half-float time with controlled variables.
PYB-01	Pencil Pressure Safety Lab (Mechanics)	Largest area protected with clear data.
PYB-02	Domino Neuron Relay (Biomedical)	Fastest reliable signal with best explanation.
PYB-03	Pulley Rescue Relay (Mechanics)	Best lift with correct setup and explanation.
PYB-04	Barcode Checksum Rescue (Computing)	Most bad codes detected with few false alarms.

THE SHARED LEARNING LOOP

Each activity uses the engineering or scientific practices that fit its actual method; fixed observation, matching, calculation, and routing tasks do not add a redesign round.

- Predict or frame the question: teams state an expectation or decision rule before the activity begins when the method calls for it.
- Build, observe, or solve: teams follow the activity-specific construction, field, microscopy, matching, calculation, or routing method.
- Measure or verify: teams follow the fixed protocol and record the required observations, calculations, or results; a baseline exists only where the activity names one.
- Score: each 100-point rubric uses the activity's named performance, evidence or reasoning, data or method, safety or cleanup, and collaboration criteria.
- Redesign where applicable: teams change one defined variable and re-test only in activities whose method and rubric include redesign.
- Defend: teams justify their final claim with specific evidence.

SAFETY OVERVIEW

No student open flames, dry ice, high voltage, hazardous chemicals, pressure vessels, dangerous projectiles, major staining, or cut-open glow sticks are used anywhere in the program.

- All electrical work is at battery voltage. Mud Battery cells are built over a tray and handled with gloves.
- Magnets are counted and controlled. Oobleck and water activities use trays and immediate cleanup.
- Pinhole optics include a no-direct-sun-view rule. Field trips follow host rules first.
- Safety controls are activity-specific: goggles for stretched slinkies; gloves and trays for soil, oobleck, or wet work; optional latex-free gloves at the ventilated polish station; and handwashing after cleanup.
- Allergy notes appear in each activity guide. The full allergy roster is confirmed before camp.

Detailed daily tasks live in the Staff Setup, Prep, and Safety Checklist.

OPERATIONS AND ROLES

- Core rule: bins packed before instruction starts, score sheets ready before work starts, and each activity's named safety controls staged before materials are distributed.
- Each team rotates roles: captain, safety lead, builder, tester, data lead, presenter.
- Best 9 of 12 primary scores count toward the final leaderboard so a single rough activity does not sink a team.
- Backups are staged and ready every day, not improvised on the spot.

MATERIALS AND PACKING BINS

Each activity has a labeled bin, with reusable and consumable items tracked per activity.

PRODUCTION NOTES

- Leaf chromatography and tie-dye were removed; Leaf Stomata Microscope Detective is used instead.
- BookBot is framed as automated storage and retrieval, bin addressing, and routing logic, not conveyor throughput.
- PY-STEM stays broad, with only selected anchor tie-ins to the Science History Institute and the BookBot.
- The two camps were audited against each other and against the legacy no-repeat list; no banned activity is reused.